



Pedro Pereira

Lisbon (Portugal)

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Senior Software Developer — Computer Graphics, Interactive Applications and Games

In 1994, after my grandparents gave me an original Game Boy and I became instantly hooked, I decided that when I grew up I'd only work on things that sparked that same sense of wonder. I've been creating games and other interactive applications since I started programming in 2009. Since then, I've been lucky to contribute to some truly exciting projects in every job I've had. Now, I'm looking forward to the next awesome challenge — somewhere I can make a real impact, working alongside talented and passionate people who once shared the same dream I did.

WORK EXPERIENCE



<https://getchaotic.com>

Lead Graphics & Gameplay Engineer | Jan 2024 – ongoing
Graphics Engineer | Jan 2023 – Dec 2023 (part-time)

- Building a high-performance 3D game engine for the Web using Rust, WebAssembly, and WebGL2
- Leading core engine architecture across graphics, gameplay, and systems engineering
- Implementing a scalable ECS architecture with real-time lighting, animation, and physics
- Developing tooling and workflows (editor integration, asset pipelines, runtime diagnostics) to improve iteration speed
- Enabling console-quality visuals and near-native performance in the browser, with a strong focus on scalability and maintainability



Unreal Engine 5 Programmer | Oct 2021 – Dec 2023

- Rebuilt the Viz Engine ↔ Unreal Engine integration, enabling Unreal's real-time rendering within Vizrt's broadcast-grade graphics platform for virtual sets and AR/VR workflows
- Enabled texture sharing and compositing for high-quality virtual graphics and real-time playout in live broadcasts
- Integrated Reality Connect to enhance presenter immersion with AI-driven depth, shadows, and reflections in virtual studio environments
- Delivered a production-ready, streamlined workflow tailored to broadcast operations, improving control and execution of complex live and augmented graphics



A BMW GROUP &
CRITICAL SOFTWARE
COMPANY

C++ Software Developer | Apr 2021 – Sep 2021

- Developed the next-generation of **BMW** navigation and infotainment systems
- Improved UX flows and navigation data presentation for safer, more intuitive driving experiences



Vision Tech Lead | May 2019 – Apr 2021

Graphics Programmer | Feb 2017 – Apr 2019

- Developed an in-house real-time rendering engine and graphical editor for TV broadcasts, virtual studios, camera-tracked, and AR content
- Led the development of custom broadcast solutions for new markets, including Virtual Stadiums, Virtual Crowds, and Virtual Ads for live sports

EDUCATION



TÉCNICO LISBOA

MSc in Information Systems and Computer Engineering

Specialization in Intelligent Systems and Games

Dissertation “Modelling Progression in Video Games”

Sep 2013 – Jun 2016 | Grade 16.8 / 20



BSc in Computer Science and Engineering

Sep 2009 – Jul 2013 | Grade 14.0 / 20

PERSONAL INTERESTS

Coding

Core Developer of “Stand Alone”, a new massive multiplayer strategy game inspired by Risk Reforged (a Warcraft 3 mod), built from scratch in **Unreal Engine 5**.

Contributing to core architecture, networking and gameplay systems, helping transition the project from a legacy mod into a modern, independent title with a focus on scalability and long-term maintainability.

Music

Active performer and former vice-president of **anTUNiA**

Tuna de Ciências e Tecnologia da Universidade Nova de Lisboa

Former member of **emCANTUS**

Adult Choir of the Associação Coral de Odivelas

Games

Recently found a new passion for sim-racing, board games and tabletop RPGs.

Started playing Dungeons & Dragons and Pathfinder, as well as painting my own miniatures and terrains.

Sports

Former practitioner of Karate, Swimming and Krav Maga.

Looking forward to training Boxing.